GG-P-681D
January 24, 1980
SUPERSEDING
Fed. Spec. GG-P-681C
October 30, 1973

FEDERAL SPECIFICATION

PROTRACTORS, CIRCULAR AND SEMICIRCULAR,

(MILS, DEGREES)

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers circular protractors of plastic and semicircular protractors of brass or plastic for mapping and precision drafting.

1.2 Classification.

1.2.1 Types, classes, styles, and sizes. Protractors shall be of the following types, classes, styles, and sizes, as specified (see 6.2):

Type I - Semicircular.
  Class A - Graduated in degrees.
  Class B - Graduated in mils.
  Style 1 - Without arc extension.
  Style 2 - With arc extension.

  Sizes (diameter).
    4-1/4 inches.
    6 inches.
    8 inches.
    10 inches.
    12 inches.

Type II - Circular, graduated in degrees.
  Style 1 - Zero to 360 degrees.
  Style 2 - Zero to 90 degree quadrants and zero to 360 degrees.

  Sizes (diameter).
    6 inches.
    8 inches.
    10 inches.

FSC 6675

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2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

Federal Specifications:

L-P-380	- Plastic Molding Material, Methacrylate.
L-P-391	- Plastic Sheets, Rods and Tubing, Rigid, Cast, Methacrylate (Multiapplication).
L-P-535	- Plastic Sheet (Sheeting), Plastic Strip; Vinyl Chloride Polymer and Vinyl Chloride-Vinyl Acetate Copolymer, Rigid.
L-P-1035	- Plastic Molding Material Vinyl Chloride Polymer and Vinyl Chloride-Vinyl Acetate Copolymer, Rigid.
QQ-B-613	- Brass, Leaded and Nonleaded: Flat Products (Plate, Bar, Sheet, and Strip).
PPP-B-636	- Boxes, Shipping, Fiberboard.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks, as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20402.

(Single copies of this specification and other Federal Specifications required by activities outside the Federal Government for bidding purposes are available without charge from Business Service Centers at the General Services Administration Regional Offices in Boston, New York, Philadelphia, Washington, DC, Atlanta, Chicago, Kansas City, MO, Fort Worth, Houston, Denver, San Francisco, Los Angeles, and Seattle, WA.

(Federal Government activities may obtain copies of Federal Specifications, Standards, and Handbooks and the Index of Federal Specifications and Standards from established distribution points in their agencies.)

Military Standards:

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.

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(Copies of Military Specifications and Standards required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless a specific issue is identified, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Associations, Inc., ATTN: Traffic Department, 1616 P Street, NW, Washington, DC 20036.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

3. REQUIREMENTS

3.1 Material. Material shall be as specified herein (see 6.4).

3.1.1 Plastics.

3.1.1.1 Molding plastics. Molding plastic shall conform to L-P-380, type I, class 3, or to L-P-1035, composition A, type III, grade GU.

3.1.1.2 Sheet plastics. Sheet plastics shall conform to L-P-391, item A, type II, grade A, or to L-P-535, composition B, type I, class 2.

3.1.2 Brass. Brass shall conform to QQ-B-613, copper alloy No. 268.

3.2 Graduations, letters, and numbers. All graduations, letters, and numbers shall be cut, stamped, printed, photoprinted, etched, or indented. Graduation lines shall be uniform, shall be between 0.004 and 0.010 inch wide, and shall have a tolerance of plus or minus five minutes of arc. Graduations, letters, and numbers shall not project above the surface of the brass or plastic. Unless otherwise specified (see 6.2), cut, stamped, etched, or indented graduations, letters, and numbers shall be filled with permanent black ink or paint.

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3.2.1 Degrees. The graduation marks for protractors reading in degrees shall be of not less than four lengths; the shortest length denoting 0.5 and the longest denoting each 10-degree increment. Graduation marks shall be numbered every 10 degrees in two rows of figures. The 0.5 degree graduation is not required on 4-1/4 inch protractors.

3.3 Crossbar. Each protractor shall have a crossbar formed by removing a semicircular segment or segments from the protractor so that the width of the crossbar and the section between the concentric arcs are equal. A vertical line shall be formed on the crossbar, intersecting the horizontal axis, to indicate the vertical axis of the protractor.

3.3.1 Horizontal axis.

3.3.1.1 Plastic protractors. On plastic protractors, the horizontal axis may be along either edge of the crossbar or within the area between the edges of the crossbar. When the horizontal axis is formed by either edge of the crossbar, a bevel not less than 1/2-inch long shall be formed on the top surface of the crossbar along the horizontal axis, centered with the line indicating the vertical axis of the protractor. The lower edge of the bevel shall be not less than 0.015-inch nor more than 0.031-inch from the bottom surface of the protractor at the point of intersection of the vertical and horizontal axis. When the horizontal axis falls within the area between the edges of the crossbar, each axis shall be represented with a line formed by the same method as that used for the graduations, and at the point of intersection of the horizontal and vertical axis a 1/16-inch diameter hole shall be drilled, countersunk or formed. The outer edge of the protractor shall be beveled on the top surface of the graduated edge.

3.4 Type I, semicircular. Type I protractors shall be of brass or plastic, as specified (see 6.2). Numbers may be slanted in the direction of increase. The outer row of numbers on the graduated arc shall read clockwise and the inner row of numbers shall read counterclockwise.

3.4.1 Brass. Brass protractors shall be flat and shall be between 1/32 and 1/16-inch thick except that 4-1/4 inch protractors shall be more than 1/64-inch, but less than 1/16-inch thick. The horizontal axis shall be along either edge of the crossbar. A vertical line shall be formed on the crossbar to indicate the vertical axis of the protractor. The line shall be formed by the same method as that used for the graduations.

3.4.2 Plastic. Plastic protractors shall be flat and shall be between 1/16 and 1/8-inch thick. Graduations shall be on the bottom surface of the plastic and shall be readily distinguishable through the thickness of plastic. Letters and numbers may be on either surface of the protractor but shall be in reading position when the graduated surface is placed in contact with the drawing, map, or other work being performed.

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3.4.3 Class A. The outer edge of the arc shall be graduated in degrees so that the zero and 180 degree marks are on the horizontal axis of the protractor. When the upper edge of the crossbar forms the horizontal axis, the graduations may be extended below that axis. The graduation marks that extend below the zero and 180 degree marks of the horizontal axis are the arc extension.

3.4.4 Class B. The outer edge of the arc shall be graduated in mils so that the zero and 3200-mil marks are on the horizontal axis of the protractor. When the upper edge of the crossbar forms the horizontal axis, the graduations may be extended below that axis.

3.4.4.1 Mils. A mil is defined as 1/6400 of a circle. The graduation marks for protractors reading mils shall be not less than three lengths; the shortest length denoting 10-mils, the medium length denoting each 50-mil increment, and the longest denoting each 100-mil increment. Graduation marks shall be numbered every 100-mils in two rows of figures.

3.4.5 Style.

3.4.5.1 Style 1. Configuration of style 1 protractors shall be similar to that shown on figure 1 (see 6.3).

3.4.5.2 Style 2. Configuration of style 2 protractors shall be similar to that shown on figure 2 (see 6.3).

3.5 Type II, circular. Type II protractors shall be of plastic. Numbers may be slanted in the direction of increase.

3.5.1 Style 1. The outer row of figures shall read clockwise from zero to 360 degrees, starting at the left in line with the horizontal axis. The inner row of figures shall read counterclockwise from zero to 360 degrees, starting at the right in line with the horizontal axis.

3.5.2 Style 2. The outer row of figures shall read clockwise from zero to 90 degrees; 90 to zero degrees; zero to 90 degrees; and 90 to zero degrees, starting at the left in line with the horizontal axis. The inner row of figures shall read clockwise from zero to 360 degrees, starting at the left in line with the horizontal axis.

3.6 Workmanship. The surface and edges of finished protractors shall be true, smooth, uniform, and polished and shall be free from defects which affect the appearance or which might impair the serviceability or use of the protractor.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. The inspection requirement specified herein are classied as follows:

(a) Quality conformance inspection (see 4.3).
(b) Inspection of packaging (see 4.5).

4.3 Quality conformance inspection.

4.3.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105, inspection level I.

4.3.2 Examination.

4.3.2.1 Samples. Samples selected in accordance with 4.3.1 shall be examined as specified in 4.4.1. AQL shall be 4.0 percent defective.

4.3.3 Tests.

4.3.3.1 Samples. Samples selected in accordance with 4.3.1 shall be tested as specified in 4.4.2. AQL shall be 4.0 percent defective.

4.4 Inspection procedure.

4.4.1 Examination. The protractors shall be examined for the following defects:

101. Type, class, size, or style not as specified.
103. Graduation, figures, and letters not as specified.
104. Dimensions not as specified.
105. Construction not as specified.
106. Workmanship not as specified.

4.4.2.1 Permanence. The permanence of the ink or paint shall be tested as follows. Moisten a clean white cloth with water and rub it along the graduated area, while exerting pressure and keeping the same portion of the cloth in contact with the protractor at all times. Rub the graduations not less than 10 times in each direction. Any effect on the graduations or any traces of coloring matter on the cloth shall constitute failure of this test.

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4.4.2.2 Accuracy. The protractor shall be measured at not less than four different intervals on the degree graduated protractor and at not less than three different intervals on the mil graduated protractor. Nonconformance to 3.2 shall constitute failure of this test.

4.5 Inspection of packaging.

4.5.1 Quality conformance inspection of pack.

4.5.1.1 Units of product. For the purpose of inspections, a completed pack prepared for shipment shall be considered a unit of product.

4.5.1.2 Sampling. Sampling for examination shall be in accordance with MIL-STD-105.

4.5.1.3 Examination. Samples selected in accordance with 4.5.1.2 shall be examined for the following defects. AQL shall be 4.0 percent defective.

107.	Materials and containers not as specified for level A. Each incorrect material or containers shall be considered one defect.
108.	Protractors of unlike description consolidated together for level A.
109.	Box for consolidation not closed and sealed as specified for level A.
110.	Protractors of unlike description packed together for level A.
111.	Marking missing, illegible, incorrect or incomplete for level A or commercial.

5. PREPARATION FOR DELIVERY

5.1 Preservation. Preservation shall be level A or commercial as specified (see 6.2).

5.1.1 Level A. Each protractor shall be contained in a plastic or paper envelope or bag and sealed.

5.1.1.1 Consolidation. Protractors of like description shall be consolidated in quantities as specified (see 6.2), in a close-fitting box conforming to PPP-B-636, WCc style optional. The box shall be closed and sealed as specified for method V in the appendix to the box specification.

5.1.2 Commercial. Protractors shall be packaged in accordance with normal commercial practice. The package shall be designed to protect the protractors against damage during shipment, handling and storage.

5.2 Packing. Packing shall be level A or commercial, as specified (see 6.2).

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5.2.1 Level A.

5.2.1.1 Military activities The protractors of like description, prepared as specified in 5.1 shall be packed together in close-fitting boxes conforming to PPP-B-636, type SF, class weather resistant. Quantities packed together shall be as specified (see 6.2), but shall not exceed a gross weight of 30 pounds. Box closure and sealing shall be in accordance with the appendix to the box specification.

5.2.1.2 Civil agencies. Packing shall be as specified in 5.2.1.1, except boxes shall be either type CF or SF.

5.2.2 Commercial. The protractors shall be packed to assure carrier acceptance and safe delivery to destination at lowest rates in fiberboard containers complying with Uniform Freight Classification rules and National Motor Freight Classification rules.

5.3 Marking.

5.3.1 Military activities. Packages and shipping containers shall be marked in accordance with MIL-STD-129.

5.3.2 Civil agencies. Marking shall be as specified in the contract or order.

6. NOTES

6.1 Intended use. The protractors are intended for use in graphically measuring angles.

6.2 Ordering data. Purchasers should select the preferred options permitted herein and include the following information in the procurement documents:

(a) Title, number, and date of this specification.
(b) Type, class, style, and size required (see 1.2).
(c) Color of filled graduations, letters, and numbers, if other than black (see 3.2).
(d) Type of material required (see 3.4).
(e) Degree of preservation and packing required (see 5.1 and 5.2).
(f) Quantity of protractors to be consolidated together (see 5.1.1.1).
(g) Quantity of protractors to be packed together (see 5.2.1.1).

6.3 Information figures. Figures 1 and 2 show styles of protractors which have been found acceptable; however, the figures are included for illustration only and are not intended to preclude the furnishing of other protractors which conform to this specification.

6.4 Recycled material. It is encouraged that recycled material be used when practical as long as it meets the requirements of the specification (see 3.1).

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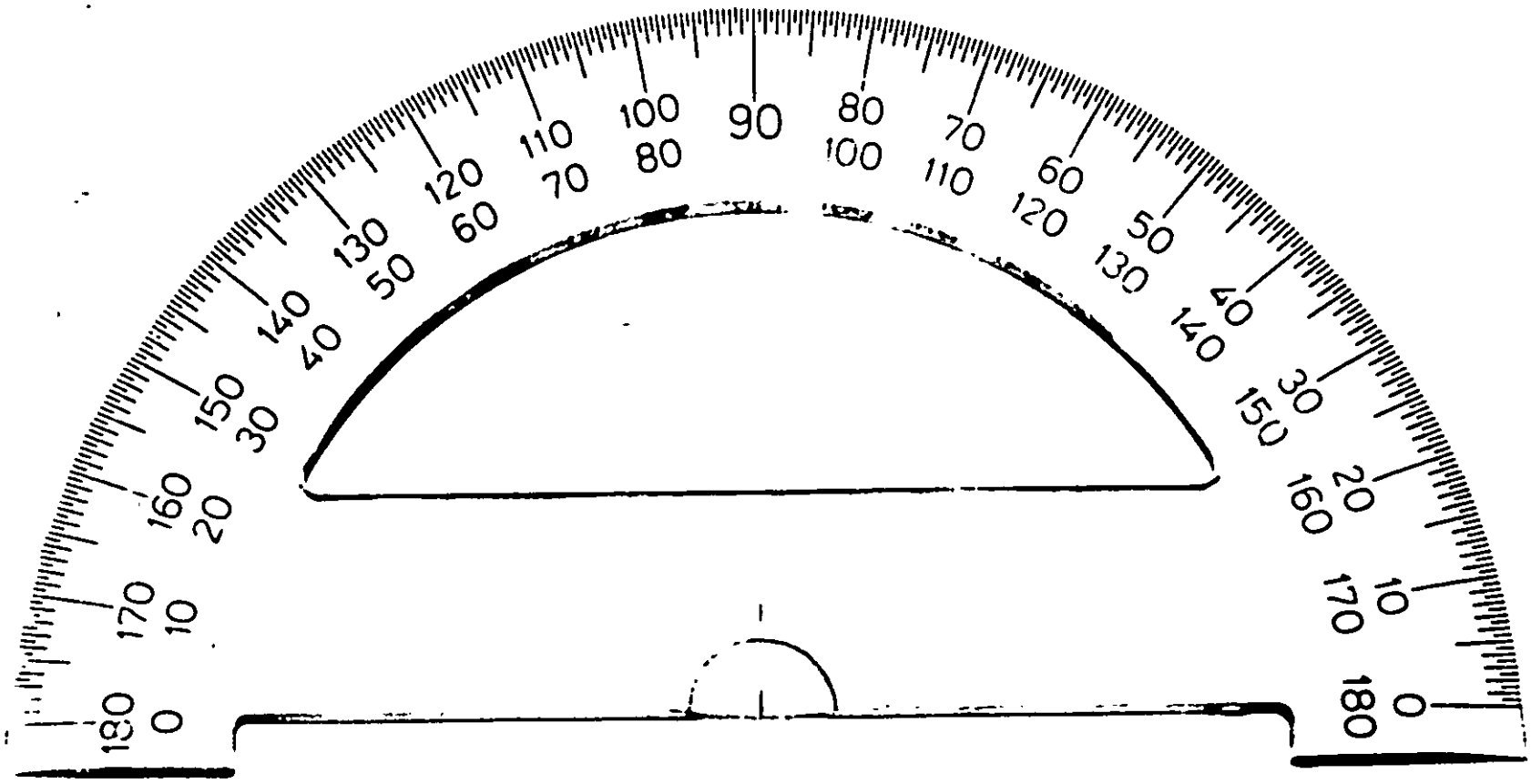


FIGURE 1. TYPE I, CLASS A, STYLE 1 PROTRACTOR WITHOUT ARC EXTENSION

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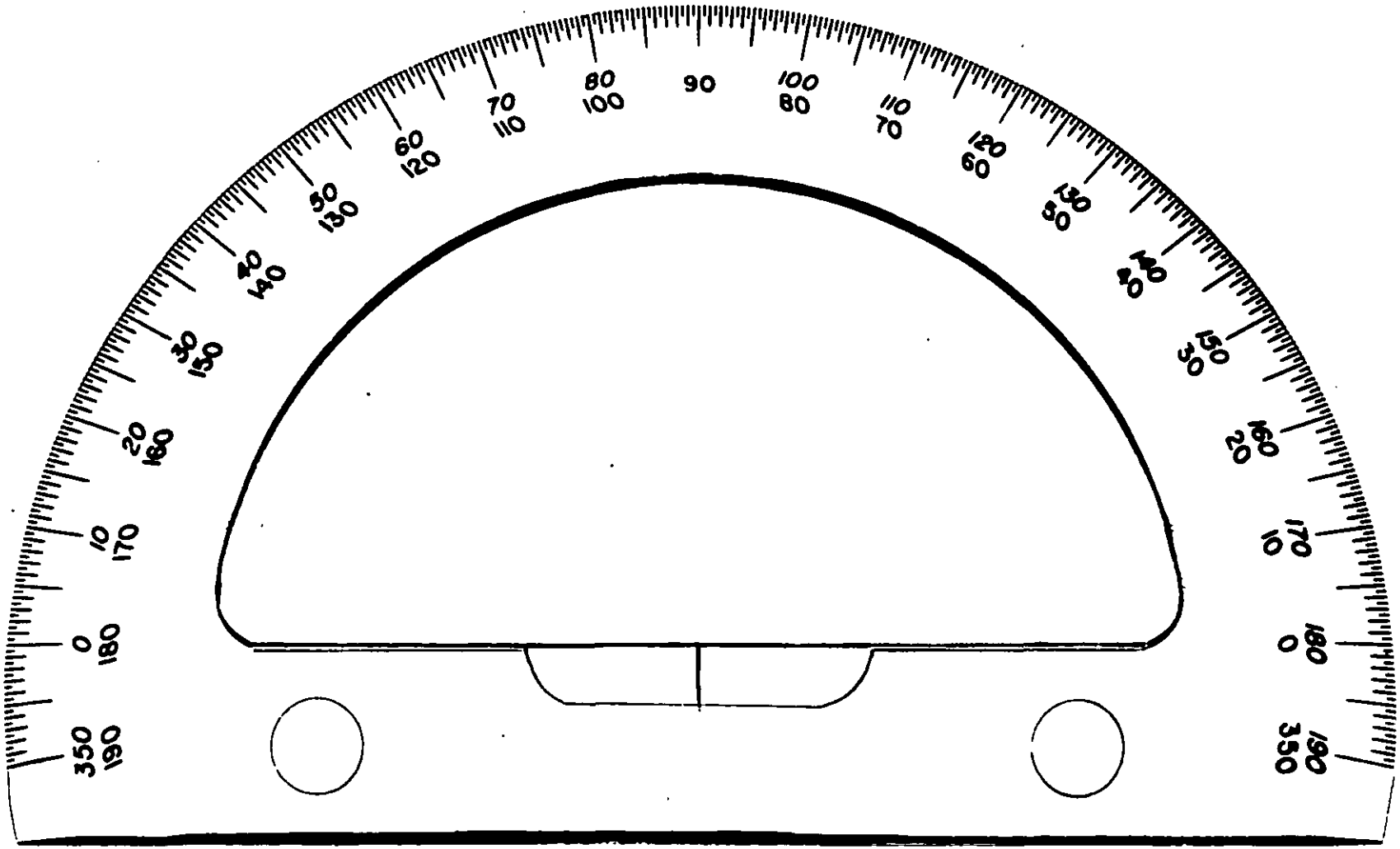


FIGURE 2. TYPE I, CLASS A, STYLE 2 PROTRACTOR WITH ARC EXTENSION

X-2408A

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MILITARY INTEREST:

Custodians:
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Review Activity:
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User Activities:
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CIVIL AGENCY COORDINATING ACTIVITY:
GSA - FSS

Preparing Activity:
Army - ME

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DOCUMENT IDENTIFIER AND TITLE
CG-P-681D Protractors, Circular and Semicircular, (Mils, Degrees)

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

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